

Project ID :

25-26J-028

1. Topic (12 words max)

Adaptive Quality of Service and Threat -Aware Response in Software-Defined Networking Using Unified Risk Classification

2. Research group the project belongs to

CI - Computing Infrastructure

3. Specialization of the project belongs to

Computer Systems and Network Engineering(CSNE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Modern Software Defined Networks (SDNs) face a dual challenge: maintaining optimal Quality of Service (QoS) and ensuring real-time threat detection. Traditional SDN-based security systems focus on identifying malicious flows, while QoS systems optimize bandwidth and delay for benign traffic. However, these two aspects are typically handled in isolation. This siloed approach often leads to inefficiencies—e.g., large but benign flows get blocked or rerouted, while small yet malicious ones go unnoticed due to low resource usage.

Previous research in QoS-aware routing or ML-based threat detection (e.g., CNN/DNN-based detection or Laplacian SVM classification) has contributed to one side of the problem. Likewise, reinforcement learning has been used to optimize flow paths for delay and congestion but rarely includes malicious behavior awareness. These disjoint systems result in wasted resources and potential vulnerabilities in production networks.

This research addresses the gap by proposing a unified model that scores each flow in real-time based on both its **Threat Score** and **QoS Score**. Threat Score measures how likely the flow is malicious, while QoS Score evaluates its impact on network resources (e.g., elephant flow behavior). Flows are then categorized in a 2x2 matrix (low/high risk × low/high QoS), enabling precise control actions such as allowing, throttling, quarantining, or dropping flows.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Our system architecture consists of four major components implemented within the SDN controller:

1. **Real-Time Traffic Monitor:** Collects flow features like byte rate, packet size, protocol, etc., from OpenFlow switches.
2. **Threat Detection Module:** Uses a lightweight ML classifier to assign a real-time **threat score** to each flow.
3. **QoS Profiling Module:** Assigns a **QoS score** based on flow volume, bandwidth use, and burst behavior.
4. **Adaptive Policy Engine:** Combines the two scores to determine and enforce appropriate actions:
 - Low threat + low QoS → allow
 - Low threat + high QoS → throttle/reroute
 - High threat + low QoS → quarantine
 - High threat + high QoS → drop

This scoring and decision model is implemented using the Ryu SDN controller, operating with real-time data from Mininet simulations. The conceptual diagram (to be attached) shows flow intake, dual scoring, decision logic, and resulting OpenFlow rule deployment.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research involves expertise in:

- Software Defined Networking (SDN) using Ryu, OpenFlow protocols
- Network Security, including behavior-based threat modeling
- Machine Learning (ML), especially traffic classification and flow scoring
- Real-time traffic monitoring and data extraction from packet flows

Key technologies include:

- Mininet for network simulation
- Ryu SDN Controller to manage and monitor flows
- Python ML for flow scoring
- Traffic datasets such as CICIDS or generated custom flows (both benign and malicious)

The system requires labeled flow data for training the threat detection ML model. This data includes statistics like flow size, duration, inter-arrival time, and protocol details. For QoS scoring, flow bandwidth and burst rates are monitored in real time.

Additionally, each module must be tuned to function within the OpenFlow rule-installation limits, requiring optimization of packet sampling and feature extraction frequency.

The model is designed for scalability and will be tested under various traffic loads to ensure real-time processing capability. Flow rules are dynamically pushed to switches using Ryu's flow mod messages.

8. Objectives and Novelty

Main Objective: To design and implement a unified, adaptive SDN framework that performs real-time flow scoring for both QoS impact and security risk, enabling intelligent threat-aware network control.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Lakshitha R.A.T.R.(IT22617514)	Real-time traffic monitoring	Design feature extractor in Ryu controller	Unified feature set for both QoS and threat scoring
Amarasinghe A.G.C.J.(IT21254048)	Threat detection engine	Train ML model to score malicious likelihood	Continuous threat scoring instead of binary detection
Mayadunne D.G.N.(IT22361172)	QoS scoring system	Implement logic to score flow bandwidth impact	Dynamic QoS score based on real-time flow metrics
Perera K.C.S.P.(IT22324306)	Adaptive decision engine	Apply flow actions based on 2D score matrix	First SDN policy engine to fuse security and performance in decision-making

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Lakshitha R.A.T.R.(IT22617514)	Aligns with Network Monitoring in SDN — collects statistical flow data and pre-processes it in controller.
Amarasinghe A.G.C.J.(IT21254048)	Aligns with Network Security — builds and applies machine learning-based behavioral threat scoring.
Mayadunne D.G.N.(IT22361172)	Aligns with QoS Engineering — models bandwidth/resource usage using live network data.
Perera K.C.S.P.(IT22324306)	Aligns with SDN Flow Management — enforces actions (block, reroute, mirror) via OpenFlow logic based on scoring.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

[illegible]

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.